

Grade 6 Accelerated
Unit 11
Lesson 9 Homework

Name _____ Answer Key _____
Date _____
Period _____

In the table, an expression has been evaluated, but the steps are mixed up. Order the expression using the numbers 1-5 on the left, the final answer has been given to you. Write an explanation for each step.

Step	Expression	Explain the Step
4	$[5]^3 \div 25$	Addition within grouping
2	$[16 \times 0.25 + 1]^3 \div 25$	Exponents within grouping
3	$[4 + 1]^3 \div 25$	Multiplication within grouping
1	$[4^2 \times 0.25 + 1]^3 \div 25$	Original expression
5	$125 \div 25$	Exponents
Answer	5	Divison

Evaluate each expression without using a calculator. Show work to support your answer.

6. $\left(\frac{1}{2}\right)^2 \times 36 + 6^2$

$$\begin{aligned} & \frac{1}{4} \times 36 + 36 \\ & 9 + 36 \\ & 45 \end{aligned}$$

7. $[(16 + 1.5)^2 - 6.25]^2 \div 100$

$$\begin{aligned} & [(17.5)^2 - 6.25]^2 \div 100 \\ & [306.25 - 6.25]^2 \div 100 \\ & [300]^2 \div 100 \\ & 90,000 \div 100 \\ & 900 \end{aligned}$$

8. $250 - (2^3 - 2)^3 \times 2$

$$\begin{aligned} & 250 - (8 - 2)^3 \times 2 \\ & 250 - (6)^3 \times 2 \\ & 250 - 216 \times 2 \\ & 250 - 432 \\ & -182 \end{aligned}$$

9. $-|-9 + \frac{1}{2}| + \frac{2}{3}(8^2 - 7^1)$

$$\begin{aligned} & -|-8.5| + \frac{2}{3}(64 - 7) \\ & -8.5 + \frac{2}{3}(57) \\ & -8.5 + 38 \\ & 29.5 \end{aligned}$$

10. $\frac{(-12.5 + |-9.25 - 4.25|)^4 + 31}{16}$

$$\begin{aligned} & \frac{(-12.5 + |-13.5|)^4 + 31}{16} \\ & \frac{(-12.5 + 13.5)^4 + 31}{16} = \frac{1^4 + 31}{16} = \frac{1 + 31}{16} = \frac{32}{16} \\ & 2 \end{aligned}$$